

第四章 静磁场

§ 4.1 静磁场及矢势

1. 恒定电流

$$\nabla \cdot \vec{J}_f(\vec{r}) = 0$$

$$\text{Ohm定律: } \vec{J}_f = \sigma \vec{E}$$

$$\text{Joule定律: } W = \int_V dV \vec{J} \cdot \vec{E} = \int_V dV \vec{E} \cdot \vec{J}_f = \int_V dV \sigma E^2$$

$$\vec{J}_f = Ne v_d = \sum_k N_k e_k \vec{v}_k \quad (\text{多种载流子})$$

导电模型:

$$eE\tau = m v_d$$

其中 τ 为平均漂移时间, v_d 为最大漂移速度.

$$\sigma = \frac{Ne v_d}{E} = \frac{Ne^2 \tau}{m}$$

在介质 ϵ, σ 中引入体分布为 ρ_f 的载流子.

$$\rho_f = \nabla \cdot \vec{D}$$

$$\frac{\partial \rho_f}{\partial t} = \nabla \cdot \frac{\partial \vec{D}}{\partial t} = \nabla \cdot (\nabla \times \vec{A} - \vec{J}_f)$$

$$= -\nabla \cdot \vec{J}_f = -\sigma \nabla \cdot \vec{E}$$

$$= -\frac{\sigma}{\epsilon} \rho_f$$

$$\Rightarrow \rho_f = \rho_{f0} e^{-\frac{\sigma}{\epsilon} t} \quad \text{衰减时间 } \tau = \frac{\epsilon}{\sigma}$$

2. 恒定电流的电场

$$\nabla \times \vec{E} = 0 \rightarrow \vec{E} = -\nabla \varphi$$

$$\nabla \cdot \vec{E} = 0 \quad \nabla \cdot \vec{J}_f = \sigma \nabla \cdot \vec{E} = -\sigma \nabla^2 \varphi = 0$$

$$\text{边界条件: } \begin{cases} \varphi_1 = \varphi_2 \\ \epsilon_1 \frac{\partial \varphi_1}{\partial n} = \epsilon_2 \frac{\partial \varphi_2}{\partial n} \end{cases}$$

3. Helmholtz 载流线圈.

圆形载流线圈

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \vec{j}(\vec{r}') dV' \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$

$$\vec{B}(z) = \frac{\mu_0}{4\pi} \frac{1}{r^2} \sin\theta \oint_C dl \times \vec{e}_z = \frac{\mu_0}{2} \frac{I a^2}{(a^2 + z^2)^{3/2}} \vec{e}_z$$

$$\frac{d\vec{B}(z)}{dz} = -\frac{3}{2} \mu_0 I a^2 \frac{z}{(a^2 + z^2)^{5/2}} \vec{e}_z$$

$$\frac{d^2\vec{B}(z)}{dz^2} = -\frac{3}{2} \mu_0 I a^2 \frac{a^2 - 4z^2}{(a^2 + z^2)^{7/2}} \vec{e}_z$$

$$\left. \frac{d^2\vec{B}(z)}{dz^2} \right|_{z=0} = 0$$

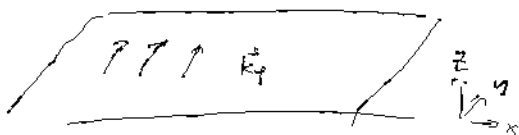
4. 载流平面的磁场

面电流 $\vec{K} = K_y \vec{e}_y$

对称性 $\vec{B} = B(z) \vec{e}_x$, $\vec{B}(z) = -B(-z)$

$$\oint \vec{B} \cdot d\vec{l} = \frac{1}{\mu_0} I$$

$$\vec{B}(\vec{r}) = \begin{cases} \frac{1}{2} \mu_0 K_y \vec{e}_x & (z > 0) \\ -\frac{1}{2} \mu_0 K_y \vec{e}_x & (z < 0) \end{cases}, \quad \vec{B}(\vec{r}) = \frac{1}{2} \mu_0 K_y \times \vec{n}$$



介质分界面的磁场边界条件.

$$\vec{B}_1 = \vec{B}_2 - \frac{1}{2} \mu_0 \vec{K} \times \vec{n}_{12}$$

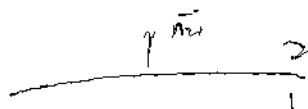
$$\vec{B}_2 = \vec{B}_1 + \frac{1}{2} \mu_0 \vec{K} \times \vec{n}_{12}$$

$$\hookrightarrow \vec{B}_{1n} = \vec{B}_{2n}$$

$$\vec{n}_{12} \times (\vec{B}_2 - \vec{B}_1) = \mu_0 \vec{K}$$

作用在边界处单位面积上磁力

$$\vec{f} = \vec{K} \times \vec{B}_S = \frac{1}{2} \vec{K} \times (\vec{B}_1 + \vec{B}_2)$$



例：对于无限长同轴传输线，已知内导线半径为 a ，电流为 I ，计算内导线受力

$$\vec{H} = \frac{I}{2\pi r} \vec{e}_\varphi, \quad \vec{B} = \frac{\mu_0 I}{2\pi r} \vec{e}_\varphi$$

$$\textcircled{1} \vec{K}_f = \vec{v} \times \vec{H} = \vec{e}_R \times \vec{H}$$

$$= \frac{I}{2\pi a} \vec{e}_z$$

$$\delta \vec{F} = \frac{1}{2} \vec{K}_f \times (\vec{B}_{in} + \vec{B}_{out}) \delta S$$

$$= -\frac{\mu_0 I^2}{8\pi^2 a^2} a d\varphi dz \vec{e}_R$$

$$\textcircled{2} \text{ 静电力密度 } \vec{T} = -\frac{1}{\mu_0} \vec{B} \vec{B} + \frac{1}{2\mu_0} B^2 \vec{e}$$

$$= -\frac{B^2}{2\mu_0} \vec{e}$$

$$\int_V \vec{T} dV = -\oint d\vec{S} \cdot \vec{T}$$

$$\delta \vec{F} = -\delta S \vec{e}_R \cdot \vec{T}$$

$$= -\frac{B^2}{2\mu_0} (\vec{e}_R \cdot \vec{e}) \delta S = -\frac{\mu_0 I^2}{8\pi^2 a} \delta\varphi \delta z$$

5. 静磁场的矢势

$$\vec{v} \times \vec{B} = \vec{j}, \quad \vec{v} \cdot \vec{B} = 0$$

$$\vec{B} \times \vec{B} = \vec{v} \times \vec{B}$$

$$\vec{B}(\vec{R}) = \frac{\mu_0}{4\pi} \int_V dV' \frac{\vec{v}' \times \vec{B}(\vec{R}')}{r} = \frac{\mu_0}{4\pi} \int_V dV' \frac{\vec{j}'(\vec{R}')}{r}$$

例：有一无限长直导线，载有电流 I ，求静矢势和磁场

$$\vec{A}(\vec{R}) = \frac{\mu_0 I}{4\pi} \int_V dV' \frac{1}{r}$$

$$= \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} dz \frac{1}{\sqrt{R^2 + z^2}} \quad z = R \sinh t$$

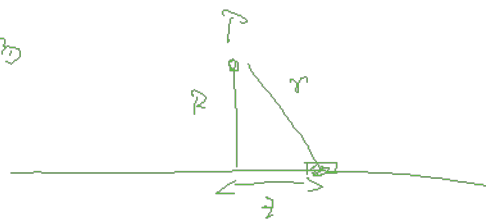
$$= \frac{\mu_0 I}{4\pi} \lim_{L \rightarrow \infty} \int_{-L}^L dz \frac{1}{\sqrt{R^2 + z^2}} = \frac{\mu_0 I}{4\pi} \int_{-L}^L dt$$

$$= \frac{\mu_0 I}{4\pi} \lim_{L \rightarrow \infty} \operatorname{arcsinh} z \Big|_{-L}^L = \frac{\mu_0 I}{4\pi} \ln(z + \sqrt{z^2 + R^2}) \Big|_{-L}^L$$

$$= \frac{\mu_0 I}{4\pi} \lim_{L \rightarrow \infty} \ln \frac{1 + \sqrt{1 + L^2/R^2}}{-1 + \sqrt{1 + L^2/R^2}} =$$

$$\triangleq A_z(R) = \frac{\mu_0 I}{4\pi} \lim_{L \rightarrow \infty} \ln \frac{1 + \sqrt{1 + L^2/R^2}}{-1 + \sqrt{1 + L^2/R^2}} - \frac{1 + \sqrt{1 + L^2/R^2}}{-1 + \sqrt{1 + L^2/R^2}}$$

$$= -\frac{\mu_0 I}{2\pi} \ln \frac{R}{R_0} \quad \text{取离导线 } R_0 \text{ 处, } A = 0$$



$$\vec{B}(\vec{r}) = \nabla \times \vec{A}(\vec{r}) = \frac{\mu_0 I}{2\pi} \frac{1}{R_0} \ln \frac{R}{R_0} \vec{e}_\phi$$

b. 矢势的规范条件.

$$\vec{A} \rightarrow \vec{A}' = \vec{A} + \nabla \psi$$

$$\vec{B}' = \nabla \times \vec{A}' = \nabla \times \vec{A} + \nabla \times (\nabla \psi) = \vec{B}$$

矢势的选取可相差任意 $\nabla \psi$

$$\int_C \vec{B} \cdot d\vec{s} = \int_C (\nabla \times \vec{A}) \cdot d\vec{s} = \oint_L \vec{A} \cdot d\vec{l}$$

矢势本身无物理意义, $\oint \vec{A} \cdot d\vec{l}$ 才有物理意义.

$$\text{Columb 规范: } \nabla \cdot \vec{A}(\vec{r}) = 0$$

例: 写出均匀磁场的矢势

$$\vec{B} = B_0 \vec{e}_z$$

$$\text{则可取 } \vec{A} = \frac{1}{2} B_0 (x \vec{e}_y - y \vec{e}_x)$$

例: 有一带电的薄球壳, 半径为 R_0 , 总电荷为 Q . 现使球壳绕自身某-轴转动

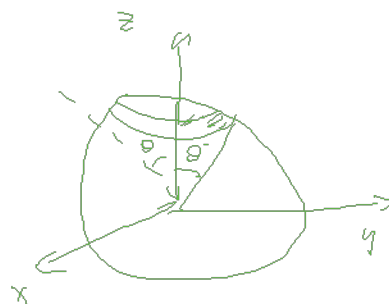
角速度为 ω . 求球壳内、外的矢势和磁场分布.

$$\text{面电流: } \vec{K}_f(\theta) = \sigma \cdot \vec{v} = \sigma \omega R_0 \sin \theta \vec{e}_\phi$$

$$\vec{r} = (0, 0, R_0),$$

$$\vec{r}' = (R_0 \sin \theta' \cos \varphi', R_0 \sin \theta' \sin \varphi', R_0 \cos \theta')$$

$$\vec{\omega} = (\omega \sin \theta, 0, \omega \cos \theta)$$



$$\begin{aligned} \vec{A}(\vec{r}) &= \frac{\mu_0}{4\pi} \int \frac{1}{|\vec{r} - \vec{r}'|} \vec{K}_f(\theta) dS' \\ &= \frac{\mu_0}{4\pi} \int \frac{1}{|\vec{r} - \vec{r}'|} \sigma \omega R_0 \sin \theta \vec{e}_\phi R_0^2 \sin \theta' d\theta' d\varphi' \\ &= \frac{\mu_0 \sigma}{4\pi} R_0^3 \int_C \frac{\vec{v}(\vec{r}')}{r} \sin \theta' d\theta' d\varphi' \end{aligned}$$

$$\vec{v}(\vec{r}') = \vec{\omega} \times \vec{r}'$$

$$= \omega R_0 (\sin \theta \sin \theta' \sin \varphi' \vec{e}_z - \sin \theta \cos \theta' \vec{e}_y + \cos \theta \sin \theta' \cos \varphi' \vec{e}_y - \cos \theta \sin \theta' \sin \varphi' \vec{e}_x)$$

$$= \omega R_0 [-\cos \theta \sin \theta' \sin \varphi', \cos \theta \sin \theta' \cos \varphi' - \sin \theta \cos \theta', \sin \theta \sin \theta' \sin \varphi']$$

$$\int_0^{2\pi} \sin \varphi d\varphi = 0, \quad \int_0^{2\pi} \cos \varphi d\varphi = 0$$

$$\rightarrow \vec{A} = \frac{\mu_0 \sigma}{4\pi} R_0^3 \int_C \frac{1}{r} \sin \theta' d\theta' d\varphi' \begin{bmatrix} -\cos \theta \sin \theta' \sin \varphi' \\ \cos \theta \sin \theta' \cos \varphi' - \sin \theta \cos \theta' \\ \sin \theta \sin \theta' \sin \varphi' \end{bmatrix}$$

$$= -\frac{1}{2} \mu_0 \epsilon R_0^3 \omega \sin \theta \int \frac{\sin \theta' \cos \theta'}{\sqrt{R^2 + R_0^2 - 2R R_0 \cos \theta'}} d\theta' \vec{e}_\theta$$

$$u = \cos \theta'$$

$$\int_{-1}^1 \frac{u du}{\sqrt{R^2 + R_0^2 - 2R R_0 u}} = -\frac{R_0^2 + R^2 + R_0 R}{3R^2 R^2} \left[\sqrt{R^2 + R_0^2 - 2R R_0 u} \right]_{-1}^{+1}$$

$$= -\frac{1}{3R^2 R^2} \left[(R_0^2 + R^2 + R_0 R) \cdot (R_0 - R) - (R_0^2 + R^2 - R_0 R) (R_0 + R) \right]$$

$$\vec{A}(R) = \frac{1}{6R^2} \mu_0 \epsilon R_0 \omega \sin \theta \left[(R_0^2 + R^2 + R_0 R) \cdot (R_0 - R) - (R_0^2 + R^2 - R_0 R) (R_0 + R) \right] \vec{e}_\theta$$

① 球壳内, $R \leq R_0$

$$\vec{A}(R) = -\frac{1}{3} \mu_0 \epsilon R_0 R \omega \sin \theta \vec{e}_\theta = \frac{\mu_0 \epsilon R_0 \omega}{3} (\vec{\omega} \times \vec{R})$$

② 球壳外

$$\vec{A}(R) = -\frac{1}{3R^2} \mu_0 \epsilon R_0^4 \omega \sin \theta \vec{e}_\theta = \frac{\mu_0 \epsilon R_0^4}{3R^2} (\vec{\omega} \times \vec{R})$$

$$\text{球壳磁矩: } m = \int \epsilon \cdot 2\pi R_0^2 \sin \theta d\theta \cdot \frac{\omega}{2\pi} \lambda R_0^2 \sin^2 \theta \vec{e}_\theta$$

$$= \frac{\omega \lambda}{3} 6\omega R_0^4 = \frac{1}{3} Q \omega R_0^2$$

$$\vec{B}(R) = \begin{cases} \frac{\mu_0}{4\pi} \frac{m}{R_0^3} (\vec{\omega} \times \vec{R}) = \frac{\mu_0}{4\pi} \frac{m}{R_0^3} \omega R \sin \theta \vec{e}_\theta & (R \leq R_0) \\ \frac{\mu_0}{4\pi} \frac{m}{R^2} (\vec{\omega} \times \vec{R}) = \frac{\mu_0}{4\pi} \frac{m}{R^2} \omega \sin \theta \vec{e}_\theta & (R > R_0) \end{cases}$$

$$\vec{B}(R) = \nabla \times \vec{A}(R) \quad \nabla = \frac{\partial}{\partial R}, \frac{1}{R} \frac{\partial}{\partial \theta}, \frac{1}{R \sin \theta} \frac{\partial}{\partial \varphi}$$

$$= \begin{cases} \frac{\mu_0}{4\pi} \frac{2m}{R_0^3} \omega (-\sin \theta \vec{e}_\theta + \cos \theta \vec{e}_R) = \frac{\mu_0}{4\pi} \frac{2m}{R_0^3} \omega \vec{e}_\theta \\ \frac{\mu_0}{4\pi} \frac{m}{R^3} \omega (2\cos \theta \vec{e}_R + \sin \theta \vec{e}_\theta) = \frac{\mu_0}{4\pi} \left[3 \frac{(\vec{m} \cdot \vec{R})}{R^5} \vec{R} - \frac{\vec{m}}{R^3} \right] \end{cases}$$

7. 麦克斯韦方程

$$\nabla \times \vec{B} = \mu_0 \nabla \times \vec{J} = \mu_0 \vec{J}$$

$$\nabla \times (\nabla \times \vec{B}) = \nabla (\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = \mu_0 \vec{J}$$

$$\text{由 } \nabla \cdot \vec{B} = 0$$

8. 矢势的值表示

$$\begin{cases} \vec{n}_i \cdot (\vec{E}_i - \vec{B}_i) = 0 & \textcircled{1} \\ \vec{n}_i \times (\vec{H}_i - \vec{E}_i) = \vec{k}_y & \textcircled{2} \end{cases}$$

$$\textcircled{1}: B_{in} = B_{out} \xrightarrow{\int_S d\vec{S} \cdot \vec{B} = \oint_L \vec{A} \cdot d\vec{l}} A_{in} = A_{out}$$

$$\vec{v} \cdot \vec{A} = 0 \Rightarrow A_{in} = A_{out}$$

$$\Rightarrow \vec{A}_i = \vec{A}_o$$

$$\textcircled{2}: \vec{n}_i \times \left[\frac{1}{R^2} (\vec{v} \times \vec{A}_i) - \frac{1}{R^2} (\vec{v} \times \vec{A}_o) \right] = \vec{k}_y$$

例：利用矢势的微分方程和边界关系，求解上例题。

$$\vec{v}^2 A_i = 0$$

$$\vec{F} = k \vec{e}_\phi = \frac{\partial u}{\partial R} \sin \theta$$

对称性：\$\vec{A} = A(R, \theta) \vec{e}_\phi\$ 轴对称性，\$\vec{v} \cdot \vec{A} = 0 \Rightarrow \vec{A}\$ 闭合。

$$\vec{v}^2 \vec{A} = \left[\frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] A(R, \theta) = 0$$

$$\begin{cases} A_{in}|_{R=R_0} = A_{out}|_{R=R_0} \\ \vec{e}_R \times \left[\vec{v} \times (A_{out}|_{R=R_0} - A_{in}|_{R=R_0}) \vec{e}_\phi \right] = \mu_0 \vec{F} \end{cases}$$

$$\vec{e}_R \times \left[-\frac{\partial}{\partial R} (A_{out}|_{R=R_0} - A_{in}|_{R=R_0}) \vec{e}_\theta + \frac{1}{R} \frac{\partial}{\partial \theta} (A_{out}|_{R=R_0} - A_{in}|_{R=R_0}) \vec{e}_R \right]$$

$$= -\frac{\partial}{\partial R} (A_{out}|_{R=R_0} - A_{in}|_{R=R_0}) \vec{e}_\phi = k \vec{e}_\phi$$

分离变量 \$A(R, \theta) = u(R) \sin \theta\$

$$\frac{1}{R^2} \frac{\partial}{\partial R} \left[R^2 \frac{\partial u}{\partial R} \right] \sin \theta + \frac{u(R)}{R^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cos \theta) - \frac{1}{R^2 \sin^2 \theta} u(R) \sin \theta = 0$$

$$\left[2R \frac{\partial u}{\partial R} + R^2 \frac{\partial^2 u}{\partial R^2} \right] \sin^2 \theta + u(R) \cos(2\theta) - u(R) = 0$$

$$R^2 \frac{\partial^2 u}{\partial R^2} + 2u \frac{\partial u}{\partial R} + 2u(R) = 0$$

$$\text{解方程 } u(R) = aR + \frac{b}{R^2}$$

$$A_{in} = aR \sin \theta, \quad A_{out} = \frac{b \sin \theta}{R^2}$$

$$\begin{cases} aR_0 = \frac{b}{R_0^2} \\ -\left(-\frac{2b}{R_0^3} - a\right) = \mu_0 k \end{cases} \quad \text{解得} \quad \begin{cases} a = \frac{1}{3} \mu_0 k \\ b = \frac{1}{3} \mu_0 k R_0^3 \end{cases}$$

$$\vec{A} = A(R, \theta) \vec{e}_\phi$$

$$= \begin{cases} \frac{1}{3} \mu_0 k R \sin \theta \vec{e}_\phi \\ \frac{1}{3} \mu_0 k \frac{R_0^3}{R^2} \sin \theta \vec{e}_\phi \end{cases}$$

9. 静磁场的唯一性定理

若 V 内给定电流和磁介质分布, $\vec{B} = \mu \vec{H}$, 并给定 ∂V 上的 $\vec{B}$ 或 $\vec{H}$ 的切向分量,

则区域内磁场唯一确定.

证: 反证, 假设存在不同 $\vec{B}$, $\vec{B}'$

$$\begin{cases} \vec{B}' = \mu \vec{H}' = \nabla \times \vec{A}' \\ \vec{B}'' = \mu \vec{H}'' = \nabla \times \vec{A}'' \end{cases}$$

$$\text{电流子} = \nabla \times \vec{H}' = \nabla \times \vec{H}''$$

$$\text{构造 } \vec{B} = \vec{B}' - \vec{B}'', \vec{H} = \vec{H}' - \vec{H}'', \vec{A} = \vec{A}' - \vec{A}''$$

$$\nabla \times \vec{H} = 0$$

$$\int_V \vec{B} \cdot \vec{H} dV = \int_V (\nabla \times \vec{A}) \cdot \vec{H} dV$$

$$= \int_V \{ \nabla \cdot (\vec{A} \times \vec{H}) + \vec{A} \cdot (\nabla \times \vec{H}) \} dV$$

$$= \int_V \nabla \cdot (\vec{A} \times \vec{H}) dV = \oint_{\partial V} \vec{A} \cdot (\vec{H} \times \vec{n})$$

$$= \oint_{\partial V} \vec{A} \cdot [(\vec{A}')_s - \vec{A}''_s] \times (\vec{H}'_s - \vec{H}''_s) = 0$$

$$\hookrightarrow \frac{1}{\mu} \int_V (\vec{B}' - \vec{B}'')^2 dV = 0$$

$$\vec{B}' = \vec{B}''$$

10. Aharonov - Bohm 效应.

$$\psi(\vec{R}) = e^{i\vec{K} \cdot \vec{R}} = e^{i\frac{\vec{p} \cdot \vec{R}}{\hbar}}$$

$$\text{正则动量 } \vec{p}_c = \vec{p} + q\vec{A} = m\vec{v} - e\vec{A}$$

$$\vec{E} = \frac{\vec{p}_c}{\hbar} = \vec{E} - \frac{e\vec{A}}{\hbar}$$

$$\text{相位差 } \Delta\phi = (\int_{C_1} - \int_{C_2}) \vec{K} \cdot d\vec{l}$$

$$= (\int_{C_1} - \int_{C_2}) \vec{E} \cdot d\vec{l} = (\int_{C_1} - \int_{C_2}) \frac{e\vec{A}}{\hbar} \cdot d\vec{l}$$

$$= \Delta\phi_0 - \frac{e}{\hbar} \oint_C \vec{A} \cdot d\vec{l}$$

$$\text{由于螺线管引起相位差 } \Delta\phi = \Delta\phi' - \Delta\phi_0 = -\frac{e}{\hbar} \oint_C \vec{A} \cdot d\vec{l} = -\frac{e}{\hbar} \Phi$$

$$\text{暗条纹: } \Delta\phi = k a b \sin \theta = (2n+1)\pi$$

$$\Delta\phi' = k\alpha b \sin\theta + \Delta\phi = (2n+1)\lambda$$

$$y \approx r \sin\theta \rightarrow r\theta = -\frac{f}{k\alpha b} \Delta\phi = \frac{e\hbar}{mV} \frac{f}{b}$$

§ 4.2 载流线圈的能量及相互作用能

1. 载流线圈静磁场总能量.

$$W = \int_{\Delta} \frac{1}{2} \vec{B} \cdot \vec{H} \, dV$$

$$\begin{aligned} \vec{B} \times \vec{H} &= (\nabla \times \vec{A}) \cdot \vec{H} = \nabla \cdot (\vec{A} \times \vec{H}) + \vec{A} \cdot (\nabla \times \vec{H}) \\ &= \nabla \cdot (\vec{A} \times \vec{H}) + \vec{A} \cdot \vec{j} \end{aligned}$$

$$\begin{aligned} W &= \oint_{\Delta} d\vec{S} \cdot (\vec{A} \times \vec{H}) + \int_V \vec{A} \cdot \vec{j} \, dV \\ &= \int_V \vec{A} \cdot \vec{j} \, dV \end{aligned}$$

形成电流分布 $\vec{j}(\vec{r})$ 和对应静磁场 $\vec{B}(\vec{r})$ 所需做功

$$\begin{aligned} \text{载流线圈 } W &= \frac{1}{2} \oint d\vec{l} \cdot \vec{A} = \frac{1}{2} \int_S d\vec{S} \cdot (\nabla \times \vec{A}) \\ &= \frac{1}{2} \oint d\vec{S} \cdot \vec{B} \\ &= \frac{1}{2} I \Phi \end{aligned}$$

考虑 δt 时间, 载流线圈内磁通量变化, 产生感应电场 $\vec{E}$ 为克服 Coulomb 力做功,

$$\begin{aligned} \delta W_{\text{ext}} &= -\delta t \oint \vec{j} \cdot \vec{E} \, d\vec{l} \\ &= -\delta t \int_V \rho \vec{j}(\vec{r}) \cdot \vec{E}(\vec{r}) \, dV \\ &= -I \delta t \oint \vec{dl} \cdot \vec{E} = -I \delta t \int_S d\vec{S} \cdot (\nabla \times \vec{E}) \\ &= I \delta t \int_S d\vec{S} \cdot \frac{\partial \vec{B}}{\partial t} = I \delta t \frac{\partial}{\partial t} \int_S d\vec{S} \cdot \vec{B} \\ &= I \delta \Phi = \delta W \end{aligned}$$

$$\begin{aligned} W &= \int \delta W \quad i = \lambda I, \delta\phi = \oint \delta\lambda \quad \lambda \in [0,1] \\ &= \int I \delta\phi = I \oint \lambda d\lambda = \frac{1}{2} I \Phi \end{aligned}$$

2. 磁相互作用能

区域 V 内分布有电流 $\vec{J}$, 区域 V_e 分布有电流 $\vec{J}_e$

$$\begin{aligned} W &= \frac{1}{2} \int_{\infty} (\vec{A} + \vec{A}_e) \cdot (\vec{J} + \vec{J}_e) dV' \\ &= \frac{1}{2} \int_{\infty} (\vec{A} \cdot \vec{J} + \vec{A}_e \cdot \vec{J}_e) dV' + \frac{1}{2} \int_{\infty} (\vec{A} \cdot \vec{J}_e + \vec{A}_e \cdot \vec{J}) dV' \\ &= W_{\infty} + V_{int} \end{aligned}$$

$$\int_{\infty} \vec{A} \cdot \vec{J}_e dV' = \int_{\infty} \vec{A}_e \cdot \vec{J} dV'$$

$$\text{证: } \vec{A}(\vec{r}) = \frac{\mu}{4\pi} \int_{V'} dV' \frac{\vec{J}(\vec{r}')}{r}, \quad \vec{A}_e(\vec{r}) = \frac{\mu}{4\pi} \int_{V_e} dV' \frac{\vec{J}_e(\vec{r}')}{r}$$

$$\begin{aligned} &\int_{\infty} \vec{A} \cdot \vec{J}_e dV' \\ &= \frac{\mu}{4\pi} \int_{\infty} dV' \left(\int_{V_e} \frac{\vec{J}_e(\vec{r}')}{|\vec{r} - \vec{r}'|} dV' \right) \cdot \vec{J}_e(\vec{r}) \end{aligned}$$

1

D.

$$\text{线圈: } V_{int} = I\phi_e = I_e\phi = \frac{1}{2}(I\phi_e + I_e\phi)$$

$$\text{其中 } \phi = \oint_{C_e} \vec{A} \cdot d\vec{l}, \quad \phi_e = \oint_{C_e} \vec{A}_e \cdot d\vec{l}$$

V_{int} 不完全由相互作用能

$$W_{int} = V_{int} - W_{\infty}$$

其中 W_{∞} 为外电源为了保持线圈中电流不变所做的功

$$\begin{aligned} W_{\infty} &= I \int (-\mathcal{E}) dt + I_e \int (-\mathcal{E}_e) dt \quad \text{抵消感生电动势所做的功} \\ &= I \int \frac{d\Phi}{dt} dt + I_e \int \frac{d\Phi_e}{dt} dt \\ &= I\Phi_e + I_e\Phi = 2V_{int} \end{aligned}$$

$$W_{int} = -V_{int} = -\frac{1}{2}(I\phi_e + I_e\phi)$$

$$\text{磁偶极子 } W = -i\Phi_e = -i\vec{m} \cdot \vec{B}_e = -\vec{m} \cdot \vec{B}_e$$

§ 4.3 静磁场磁标势描述

1. 磁标势描述

若无传导电流, $\vec{j}_f = 0$: $\nabla \times \vec{H} = 0$

此时有 Ampère 环路定理 $\oint_C d\vec{l} \cdot \vec{H} = \int_S d\vec{S} \cdot \vec{j}$

引入标势 $\vec{H} = -\nabla \varphi_m$

代入 Ampère 环路定理:

$$\oint_C d\vec{l} \cdot \vec{H} = \oint_C d\vec{l} \cdot (\nabla \times \vec{H}) = - \int_S d\vec{S} \cdot (\nabla \times \nabla \varphi_m) = 0$$

$$= \oint_{L_1} d\vec{l} \cdot \vec{H} = \oint_{S_1} d\vec{S} \cdot \vec{j} = 0$$

$$= \oint_{L_2} d\vec{l} \cdot \vec{H} = \oint_{S_2} d\vec{S} \cdot \vec{j} \neq 0$$



以载流线圈为分界面划分区域

$$\text{区域内 } \vec{H} = -\nabla \varphi_m, \quad \vec{B} = -\frac{1}{\mu} \nabla \varphi_m$$

2. 磁标势微分方程.

$$\vec{B} = \mu_0 (\vec{H} + \vec{M}), \quad \nabla \cdot \vec{B} = 0$$

$$\hookrightarrow \nabla \cdot \vec{H} = -\nabla \cdot \vec{M}$$

$$\text{定义 } \rho_m = -\mu_0 \nabla \cdot \vec{M} \quad \rho_f = -\nabla \cdot \vec{P}$$

$$\nabla \cdot \vec{H} = \frac{\rho_m}{\mu_0}$$

$$\nabla^2 \varphi_m = -\frac{\rho_m}{\mu_0}$$

3. 磁标势边值关系

$$\int \vec{n}_i \times (\vec{H}_i - \vec{H}_e) = \vec{K}_f \quad ①$$

$$\vec{n}_i \cdot (\vec{B}_i - \vec{B}_e) = 0 \quad ②$$

①: 若 $\vec{K}_f = 0$, $H_{te} = H_{ti}$, $\varphi_i = \varphi_e$, 磁标势连续

$$\textcircled{2}: B_n = \vec{n} \cdot (\vec{H}_i + \vec{M}_i) = \vec{n} \cdot (\vec{H}_e + \vec{M}_e)$$

$$\frac{\partial \varphi_i}{\partial n} - \frac{\partial \varphi_e}{\partial n} = \vec{n} \cdot (\vec{M}_e - \vec{M}_i)$$

$$\vec{B} = \mu \vec{H}$$

$$\mu \frac{\partial \psi_1}{\partial n} = \mu_0 \frac{\partial \psi_0}{\partial n}$$

§ 4.4. 矢势磁多极展开.

1. 矢势多极展开及磁偶极矩.

$$\vec{A}(\vec{R}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{R}') dV'}{r} = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{R}') dV'}{|\vec{R} - \vec{R}'|}$$

$$\text{展开: } \frac{1}{|\vec{R} - \vec{R}'|} = \frac{1}{R} - \vec{R}' \cdot \nabla \frac{1}{R} + \frac{1}{2} \vec{R}' \vec{R}': \nabla \nabla \frac{1}{R}$$

$$\begin{aligned} \vec{A}(\vec{R}) &= \frac{\mu_0}{4\pi} \int_V dV' \vec{j} \frac{1}{R} - \frac{\mu_0}{4\pi} \int_V dV' \vec{j} \cdot \vec{R}' \nabla \frac{1}{R} \\ &\quad + \frac{\mu_0}{8\pi} \int_V dV' \vec{j} \cdot \vec{R}' \vec{R}' : \nabla \nabla \frac{1}{R} + \dots \\ &= \vec{A}^{(0)} + \vec{A}^{(1)} + \vec{A}^{(2)} + \dots \end{aligned}$$

矢势场 $\vec{j}$ 在 V 的边界面上满足 $\vec{n} \cdot \vec{j} = 0$,

$$\text{则 } \int_V \vec{j} dV = - \int_V dV (\nabla \cdot \vec{j}) \vec{r}$$

$$\int_V dV \vec{j} \cdot \vec{r} + \int_V dV \vec{r} \cdot \vec{j} = - \int_V dV (\nabla \cdot \vec{j}) \vec{r} \cdot \vec{r}$$

$$\vec{n} \cdot \vec{j} = 0, \quad \nabla \cdot \vec{j} = 0$$

$$\therefore \begin{cases} \int_V \vec{j}(\vec{R}') dV' = 0 \\ \int_V (\vec{j} \cdot \vec{R}' + \vec{R}' \cdot \vec{j}) dV' = 0 \end{cases}$$

$$\vec{A}^{(0)} = 0$$

$$\begin{aligned} \vec{A}^{(1)} &= - \frac{\mu_0}{4\pi} \int_V dV' \vec{j} (\vec{R}' \cdot \nabla \frac{1}{R}) \\ &= - \frac{\mu_0}{4\pi} \int_V dV' (\vec{j} \cdot \vec{R}') \cdot \nabla \frac{1}{R} \\ &= \frac{\mu_0}{8\pi} \int_V dV' (\vec{j} \cdot \vec{R}' - \vec{R}' \cdot \vec{j}) \cdot \frac{\vec{R}}{R^3} \quad \nabla \frac{1}{R} = - \frac{\vec{R}}{R^3} \\ &= \frac{\mu_0}{8\pi} \int_V dV' (\vec{R}' \times \vec{j}) \times \frac{\vec{R}}{R^3} \end{aligned}$$

$$\begin{aligned} (\vec{R}' \times \vec{j}) \times \vec{R} &= \vec{R} \times (\vec{j} \times \vec{R}') \\ &= \vec{j} (\vec{R}' \cdot \vec{R}) - \vec{R}' (\vec{j} \cdot \vec{R}) \\ &= (\vec{j} \cdot \vec{R}' - \vec{R}' \cdot \vec{j}) \cdot \vec{R} \end{aligned}$$

$$\vec{A}^{(1)} = \frac{\mu_0}{4\pi} \vec{m} \times \frac{\vec{R}}{R^3} \quad \text{其中 } \vec{m} = \frac{1}{2} \int_V \vec{R}' \times \vec{j} dV' \text{ 为磁偶极矩.}$$

2. 平面载流线圈的磁场.

$$\vec{m} = \frac{1}{2} \oint \vec{r}' \times I d\vec{l}' = I \vec{S}$$

3. 载流线圈在远场的磁场

$$\begin{aligned} \vec{B} &\approx \nabla \times \vec{A}^{(1)} = \frac{\mu_0}{4\pi} \nabla \times \left(\vec{m} \times \frac{\vec{R}}{R^3} \right) = \nabla \times \left(\vec{m} \times \frac{\vec{R}}{R^3} \right) \\ &= -\frac{\mu_0}{4\pi} (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} = \left\langle \frac{\vec{R}}{R^3} \cdot \nabla \right\rangle \cdot \vec{m} - (\nabla \cdot \vec{m}) \frac{\vec{R}}{R^3} + \left(\nabla \cdot \frac{\vec{R}}{R^3} \right) \vec{m} - \langle \vec{m} \cdot \nabla \rangle \frac{\vec{R}}{R^3} \\ &= -\frac{\mu_0}{4\pi} \nabla \left(\vec{m} \cdot \frac{\vec{R}}{R^3} \right) = -(\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} \\ &= -\mu_0 \nabla \varphi_m \\ \varphi_m &\approx \frac{\vec{m} \cdot \vec{R}}{4\pi R^3} \\ &= \vec{m} \times (\nabla \times \frac{\vec{R}}{R^3}) + (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} = (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} \end{aligned}$$

4. 点磁偶极子的矢势、磁场及电流分布

$$\begin{aligned} \vec{A} &= \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{R}}{R^3}, \quad \varphi_m = \frac{\vec{m} \cdot \vec{R}}{4\pi R^3} \\ \vec{B} &= \nabla \times \vec{A} = \frac{\mu_0}{4\pi} \nabla \times \left(\vec{m} \times \frac{\vec{R}}{R^3} \right) \\ &= \frac{\mu_0}{4\pi} \left[\vec{m} \nabla \cdot \frac{\vec{R}}{R^3} - (\vec{m} \cdot \nabla) \frac{\vec{R}}{R^3} \right] \\ &= \mu_0 \vec{m} \delta(\vec{R}) - \frac{\mu_0}{4\pi} \nabla \left(\vec{m} \cdot \frac{\vec{R}}{R^3} \right) \end{aligned}$$

$$\begin{aligned} \text{非原点: } \vec{B} &= -\frac{\mu_0}{4\pi} \nabla \left(\vec{m} \cdot \frac{\vec{R}}{R^3} \right) \\ &= -\frac{\mu_0}{4\pi} \left[\frac{\vec{m}}{R^3} - 3 \frac{(\vec{m} \cdot \vec{R}) \vec{R}}{R^5} \right] \end{aligned}$$

$$\begin{aligned} \text{电流: } \vec{J}_g &= \frac{1}{\mu_0} \nabla \times \vec{B} \\ &= \frac{1}{\mu_0} \nabla \times \mu_0 \vec{m} \delta(\vec{R}) - \frac{1}{4\pi} \nabla \times \nabla \left(\vec{m} \cdot \frac{\vec{R}}{R^3} \right) \\ &= \vec{m} \times \nabla \delta(\vec{R}) \\ &= -\vec{m} \times \nabla \delta(\vec{R}) \end{aligned}$$

5. 点电偶极子层

$$\begin{aligned} \vec{A} &= \frac{\mu_0}{4\pi} \int_S d\vec{l}' \times \frac{(\vec{R} - \vec{R}')}{|\vec{R} - \vec{R}'|^3} = \frac{\mu_0}{4\pi} I \int_S d\vec{l}' \times \sigma' \frac{1}{|\vec{R} - \vec{R}'|} \\ &= \frac{\mu_0 I}{4\pi} \int_S \frac{d\vec{l}'}{|\vec{R} - \vec{R}'|} \quad \oint d\vec{l}' \times \sigma \varphi(\vec{R}') = \oint_C d\vec{l} \varphi(\vec{R}) \end{aligned}$$

$$\begin{aligned} \varphi_m &= \int d\varphi_m = \int \frac{\vec{m} \cdot \vec{R}}{4\pi R^3} \\ &= \frac{I}{\mu_0} \int d\vec{l}' \cdot \frac{\vec{R}}{R^3} = \frac{I}{\mu_0} \int d\Omega \end{aligned}$$

$$1 = \frac{2}{47} \Delta \Omega$$